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DS 5030

Dr. Johnson

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Lab 7 Write-up

Summary of the Problem

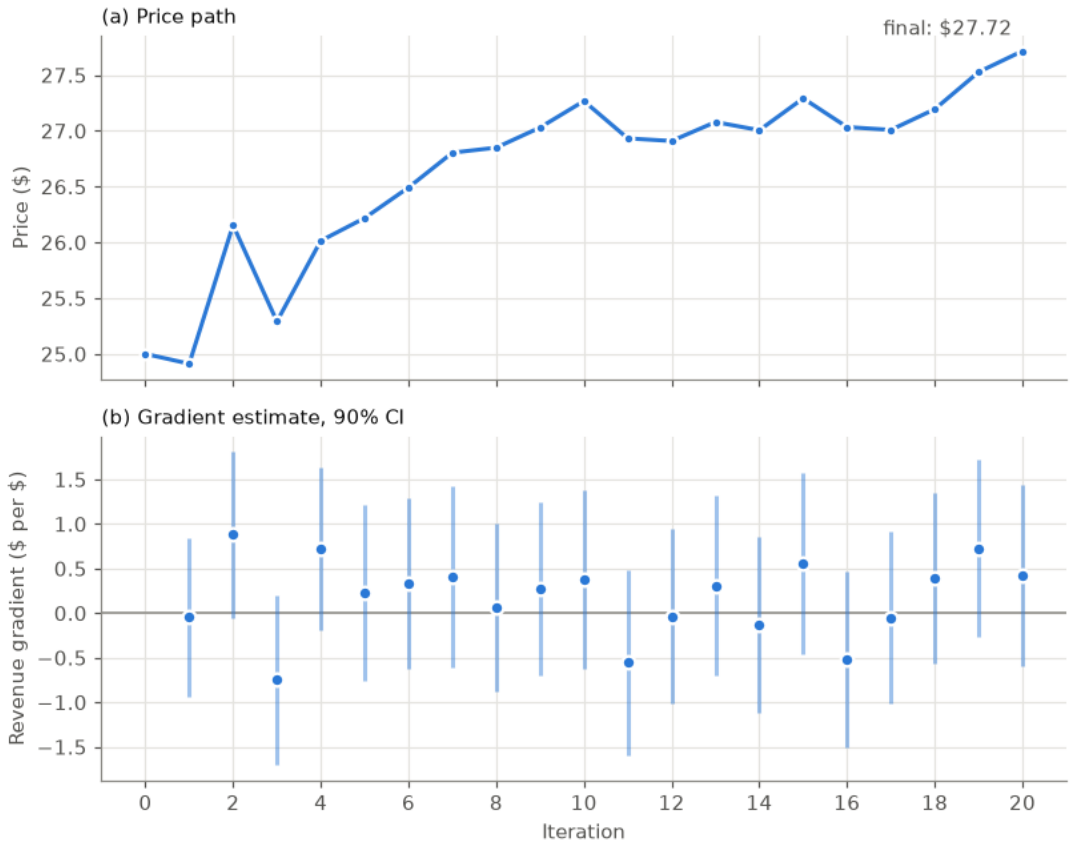
The firm has to choose a price for its product without knowing how demand responds to price. The firm must offer a price to a randomly arriving set of customers and see who buys. We know the revenue-maximizing price is somewhere between $0 and $70. The revenue curve isn't smooth or single-peaked, so several prices can look optimal while only one is the global optimum. Every experiment also has a cost. Any customer who sees a price other than the optimal one earns us less than the $11.925 expected profit they would bring at the optimum, and that lost profit adds up quickly. The goal is to design an experiment that reliably finds the globally optimal price while losing as little profit as possible. We have to balance having enough prices to avoid getting stuck on a local peak and moving customers to good prices fast enough to keep the cost low.

Uncertainty

The number of customer arrivals each day varies, and the probability of a purchase depends on the price point, making both their arrivals and purchases random. This causes revenue values at each price point to be uncertain. We are also uncertain about measuring the effect each change in price has on revenue, since there is a possibility the change is due to random chance rather than an actual difference. Our price selection method accounts for the randomness by determining statistical significance between price points and ensuring that we arrive at the optimal price to maximize revenue.

We first use a grid search on prices $5 to $65 with an interval of $10 and compare the average revenue per customer at two different prices and take the difference. We create a 95% confidence interval for that difference to determine if the two price points generate significantly different revenue amounts. If the confidence interval does not contain zero, we know that one price generates more revenue than another, and thus the losing price point is eliminated from comparison. If the confidence interval contains zero, we examine both price points, as either price could bring us the closest to maximizing revenue. After determining that a price of $25 dominates all other prices through grid search, we begin our gradient search there and adjust it over 20 iterations to find the maximum revenue. Our learning rate begins at 2 and decreases each iteration with the

formula $\frac{2}{\sqrt{iteration+1}}$, ensuring each price adjustment is not too large and we are able to find the optimal price of a bowl, $27.72.



Challenges

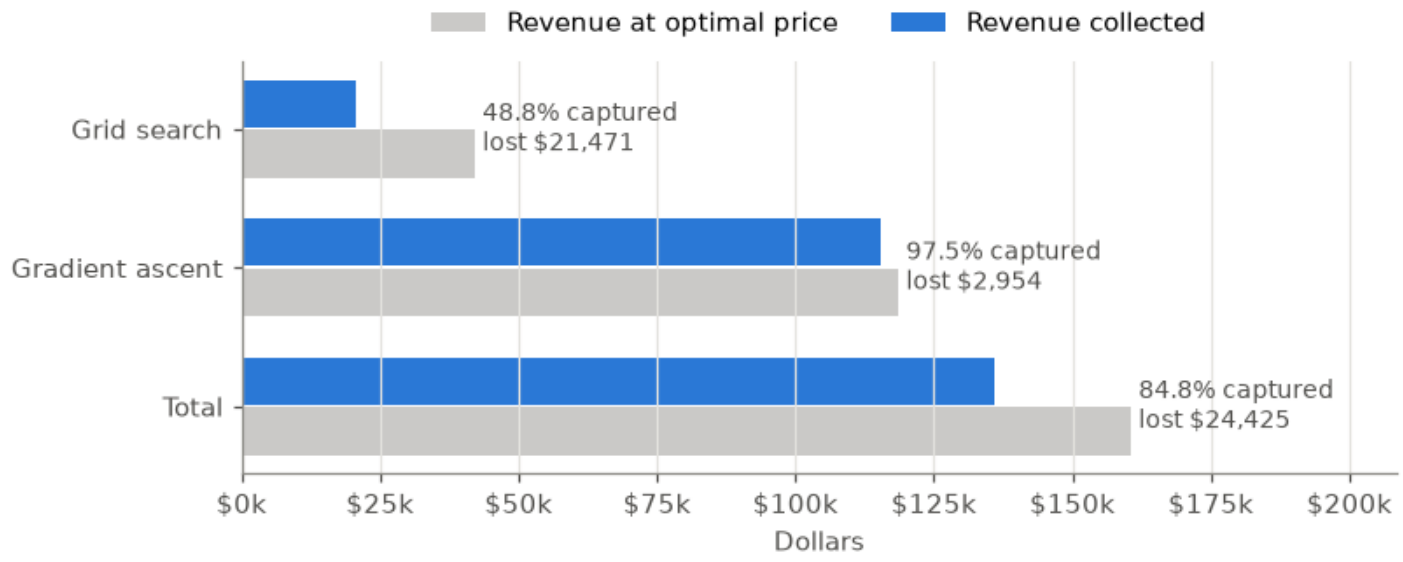
During this lab, we face multiple challenges when creating a process to find the optimal price of the firm's product and when creating the code to complete the calculations. Although we were given most of the code to complete this task, fine tuning the processes was not straight forward to minimize revenue loss was difficult. Testing every single price point to using grid search would be a great way to find the product's optimal price but it would cost too much money. Also, we cannot use the gradient search because there are multiple local peaks in that gradient search may mistake as the global optimal price. To solve this problem, we used grid search to find the general range of where the global optimal price is located. With our limited budget, we wanted to test as little prices as possible but still get a good estimate on the general location of the global optimal price. We determined that testing the prices from $5 to $65 with an interval of $10 with roughly 500 customers per price point was a good way to get a general sense of the location of the

global optimal price. After determining the global optimal price was around $25, we used gradient search to hunt for the optimal price for the product at around $25. The two searches cost the firm a reasonable $24,424.69 in lost revenue, solving our first problem of finding the global optimal price without blowing the firm's budget

Our second problem was creating a way to calculate lost revenue into our code. To solve this problem, we created code that compares the revenue made at each sampled price point to the optimal amount of revenue that could have been made if we sold each product for its optimal price from the very beginning. To make things a little easier, the lab provided us with the optimal profit for each transaction, $11.925. By subtracting the amount of revenue that could have been made at optimal price minus the revenue made at the product's sample price, we were able to get the opportunity cost of not pricing the product at the optimal price point which we called "cost" for simplicity reasons. We used this technique for all the different price samples in the grid search and gradient search. The optimal revenues and sampled revenues were calculated for both searches separately to allow us to see what search had the highest cost. At the end of the lab, we combined all the figures to present our grad total for lost revenue of $24,424.69.

4. One paragraph that summarizes your results. In particular, how many consumers did you experiment on, what were your revenues during the experiments, and what percentage of the optimal profit ($11.925 per customer) did you capture?

4. Summary of Results



We found the best price in two steps. First, we tried seven widely spaced prices, from $5 to $65, on about 3,500 customers to see which part of the price range worked best. $25 clearly came out ahead. This step was costly, because many of those customers saw prices we now know were poor, and they brought in only $5.83 each on average. Second, we made small price adjustments around $25, nudging the price up or down based on how sales responded, on about 9,900 more customers. Those customers were already seeing good prices, so they brought in $11.63 each which is nearly optimal. The search settled on a price of $27.72. Altogether we tested on 13,452 customers and earned $135,990, or $10.11 per customer. If every one of them been charged the best price from the start, each would have earned us $11.925. The $1.81 difference per customer is the cost of learning which price works, about $24,400 in total, and most of it came from the grid search. That means we kept about 85% of the best possible profit while running the experiment. Going forward, charging $27.72 would earn very close to the full $11.925 per customer.